

Section: **3.5 – STRENGTH OF COMPOSITE SANDWICH PANELS**

Chapter: **3530**

Title: **RECTANGULAR SANDWICH PANELS**

Revision: **0.1**

Date: **18/06/2018**

1 INTRODUCTION

1.1 OBJECT

This chapter is dedicated to describing the basics of a methodology to calculate the displacements and internal loads distribution of a simply supported rectangular sandwich panel subjected to pressure and in plane loads, as well as its buckling onset, taking into account not only the membrane and bending problem (thin plates) but the potential contribution of transverse shear.

For other geometries or more complex loads states the use of dedicated FEM models for obtaining the internal loads is recommended.

1.2 SCOPE AND LIMITATIONS

Internals loads and buckling analysis of rectangular plane sandwich panels under the following assumptions (Ref. 1):

- ✓ Core contribution to bending stiffness is negligible and transverse shear stress is constant through its thickness
- ✓ Core is rigid in transversal direction
- ✓ Transversal deflections are small (no “stretching” of the middle plane)
- ✓ Sandwich faces are thin compared with core height ($d/t > 6$)
- ✓ Core is orthotropic and characterized by its shear moduli G_{xz} and G_{yz}

Other geometries or complex loads states shall be treated by other means (typically numerical FEM analysis).

2 METHOD DESCRIPTION

A rectangular plane panel is considered, of length a in direction X and b in direction Y , subjected to N_x , N_y and N_{xy} forces in the middle plane, and a pressure p . The sandwich mid-plane contains the reference plane XY . The displacement of the panel in Z direction is noted by w in further discussion.

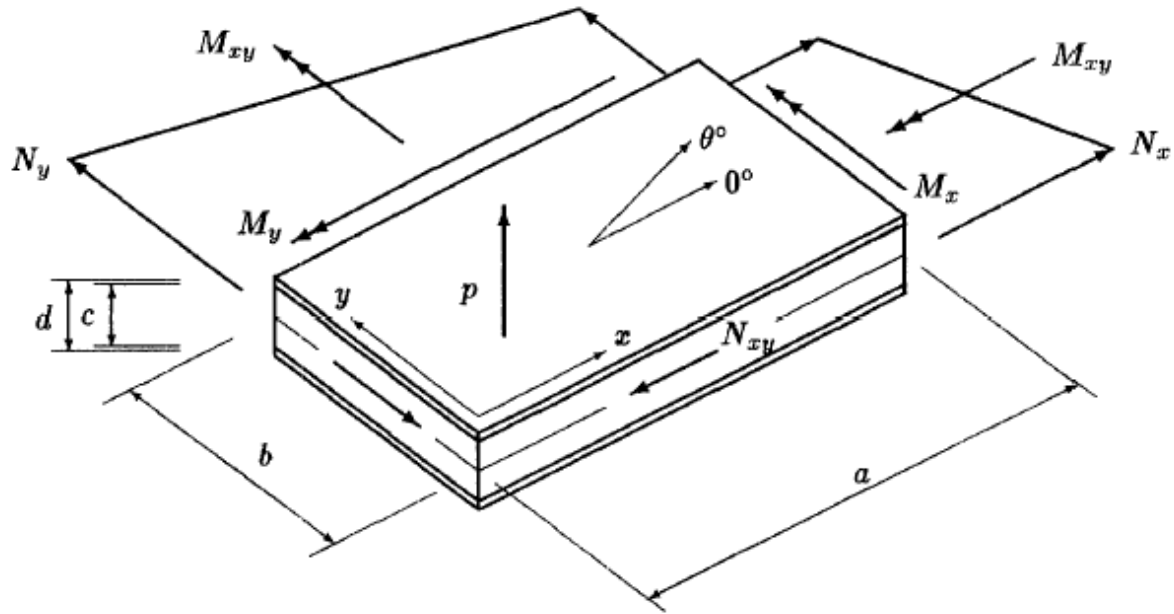


Figure 1. Reference axis system and signs conventions

Classical lamination theory can be extended to compute transverse shear effects using, for example Libove and Batdorf theory, as discussed in Ref. 1 (not all details presented herein).

$$\{M\} = \begin{Bmatrix} M_x \\ M_y \\ M_{xy} \end{Bmatrix} \quad \{K_s\} = \begin{Bmatrix} -\left(\frac{\partial^2 w}{\partial x^2} - \frac{1}{S_x} \frac{\partial Q_x}{\partial x}\right) \\ -\left(\frac{\partial^2 w}{\partial y^2} - \frac{1}{S_y} \frac{\partial Q_y}{\partial y}\right) \\ -\left(2\frac{\partial^2 w}{\partial x \partial y} - \frac{1}{S_x} \frac{\partial Q_x}{\partial y} - \frac{1}{S_y} \frac{\partial Q_y}{\partial x}\right) \end{Bmatrix}$$

$$S_x = G_{zx} \frac{d^2}{c} \quad S_y = G_{yz} \frac{d^2}{c}$$

$$\{M\} = ([D] - [B][A]^{-1}[B]) \{K_s\}$$

Total energy of the system Π shall be the addition of the deformation energy of the sandwich panel U plus the potential energy of the applied loads V :

$$\Pi = U + V$$

- ✓ Deformation energy

$$U = \frac{1}{2} \int_0^b \int_0^a \{M\}' \{K_s\} dx dy + \frac{1}{2} \int_0^b \int_0^a \left(\frac{Q_x^2}{S_x} + \frac{Q_y^2}{S_y} \right) dx dy$$

- ✓ Potential energy of applied loads

$$V = \frac{1}{2} \int_0^b \int_0^a \left[N_x \left(\frac{\partial w}{\partial x} \right)^2 + N_y \left(\frac{\partial w}{\partial y} \right)^2 + 2N_{xy} \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \right] dx dy + \\ + \int_0^b \int_0^a \{N\}' [A]^{-1} [B] \{K_s\} dx dy - \int_0^b \int_0^a p w dx dy$$

Rayleigh-Ritz method can be used to obtain the solution by approximating the unknowns (w , Q_x and Q_y distributions) by series of shape functions compliant with boundary conditions, with undetermined amplitude coefficients, and then searching for the minimum of the total energy. For a simply supported panel the following shape functions can be selected and boundary conditions imposed.

$$w = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} A_{ij} \sin(\alpha_i x) \sin(\beta_j y)$$

$$\alpha_i = \frac{i\pi}{a}$$

$$w = Q_y = M_x = 0 \\ \dots \text{at } x = 0 \text{ and } x = a$$

$$Q_x = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} B_{ij} \cos(\alpha_i x) \sin(\beta_j y)$$

$$\beta_j = \frac{j\pi}{b}$$

$$w = Q_x = M_y = 0 \\ \dots \text{at } y = 0 \text{ and } y = b$$

$$Q_y = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} C_{ij} \sin(\alpha_i x) \cos(\beta_j y)$$

From these initial equations, it is possible to reach a linear system of equations to solve the unknown coefficients A_{ij} and the rest of magnitudes of interest, including displacement w and internal loads. The complete formulation will not be shown here.

On the other hand, the eigenvalues of that linear system of equations shall correspond to the factors to apply to applied loads to reach the buckling loss of stability (i.e. the buckling levels) for the different buckling modes.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Length of a rectangular panel in direction X
[A]	N/mm	Stiffness matrix of a composite laminate in in-plane (membrane) loads
b	mm	Width of a rectangular panel
[B]	N	Stiffness matrix of a composite laminate determining the membrane/bending coupling
c	mm / -	Core thickness or, as subscript, relative to the core of the sandwich panel
d	mm	Distance between face sheet centroids
[D]	Nmm	Stiffness matrix of a composite laminate in out/plane (membrane & torsion) moments
G	MPa	Shear modulus
M	Nmm/mm	Shell out-plane (bending and torsion) moments
N	N/mm	Shell in-plane (membrane) loads
Q	N/mm	Shell transverse shear load
p	MPa	Pressure on a panel
X, Y, Z	-	Direction of a local reference axis system, with X and Y parallel to sandwich plane, and X aligned with material reference 0° direction
U	Nmm ²	Deformation energy of the sandwich
V	Nmm ²	Potential energy of applied loads
Π	Nmm ²	Total energy of the system
w	mm	Displacement of the middle plane in transversal direction (Z axis)
K _s	mm ⁻¹	Curvature deformation extended with transverse shear

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	NT-T-ADP-0005	Ed. A	MANUAL TEORICO DEL PROGRAMA PANSA
Ref. 2	[SAE] CMH-17	2013	COMPOSITE MATERIALS HANDBOOK. VOLUME 6. STRUCTURAL SANDWICH COMPOSITES.
Ref. 3	[Robert M. Jones]	2 nd Ed.	MECHANICS OF COMPOSITE MATERIALS
Ref. 4	[S. Ward, L. Gintert]	2001	Analysis of Sandwich Structures
Ref. 5	[C. Kassapoglou]	2 nd Ed.	Design and Analysis of Composite Structures

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures
PANSA (Airbus DS)	5.5	Stress, strength and buckling analysis of composite sandwich panels

Table 1. Software programs referenced in this chapter

DISTRIBUTION LIST

<i>Department</i>	<i>Name</i>	<i>Full Doc.</i>	<i>Cover Page</i>
Stress Methods & Optimisation (TEASP-TL3)	Dr. Fernaß Daoud	X	
Stress Methods & Optimisation (TEASP-TL3)	Dr. Alexander Hörmann	X	
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
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Structural Repair GET (TEASA-TL6)	Jose Luis Melchor Martin de la Sierra	X	

Table 2. Distribution list (of electronic document, ahead of public release)

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
0.1 18/06/2018	G. Baños	First issue	All pages

Table 3. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Analysis (TEASA)	Name DD/MM/2018	Name DD/MM/2018
Stress Methods (TEASP-TL3)	G. Baños 18/06/2018	Name DD/MM/2018

Table 4. Approval signatures table for last revision of this chapter